

Manual

Complaint Management REK-EVA 8.1

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1 Complaint Management - Overview

Fields of application

This component enables the creation of complaints and complaint details including the assignment of measures and the performance/documentation of the failure analysis.

Implementation notes

This component is recommended if you wish to record complaints by customers, suppliers or internal complaints.

Integration

This component mainly refers to the components

- Complaints analysis
- Analysis of complaint costs
- Failure analysis of complaints and
- Tracing of measures

but is, to some extent, also combined with other components.

Features

These functions are provided.

- Editing function for the creation and modification of relevant master data (articles, companies, failures, measures, cost types, parties in charge, persons, etc.)
- Distinction of different complaint types, e.g. complaints by customers, suppliers as well as internal complaints
- Differentiation between header and partial complaint data including the possibility to create n partial complaints for one complaint header
- Assignment of different statuses (e.g. recorded, in process, completed) and results (justified (acknowledged), unjustified (rejected), goodwill, warranty) separated by the complaint header and the assigned partial complaints
- Assignment of measures of different types (short-term, medium-term, long-term) including editing status and specification of dates, degree of fulfillment, effectiveness as well as responsibilities (parties in charge)
- Assignment of the relevant inspection order
- Detailed failure analysis by the failure tree including assignment of failure types, failure locations, failure causes, causers, measures, documents, defect characteristics

2 Summary

2.1 Notes on the Document

This document describes the “article” application of the Manufacturing Operation Center (MOC). General information on how to use MOC can be found in the document entitled “[moc_cc.pdf](#)”.

3 Article

The article catalog has been designed to edit/keep articles. Article data is a global catalog that is used in many CAQ modules and in PDV (Process Data Collection). Provided that there is an interface to a higher-level system (e.g. ERP system), articles may be created automatically via this interface. As soon as a new article is created or changed, e.g. in the ERP system, the article data record is automatically created or changed in the HYDRA-CAQ article catalog based on the defined information.

3.1 Starting the Function

Menu	Master data → Quality management → Article Master data → Process data processing → Article
Transaction code	atc
Function authorization	atc

3.2 Default Application Layout

3.3 Toolbar

The toolbar provides the different functions available for this application and possibly links to other applications. The functions included in the “general” tab of the toolbar refer to all detail applications. In addition to the standard functions, such as help, request data, save application settings, and print preview, the other tabs also include specific functions that are specially tailored to the respective detail application. The following sections describe the individual application functions.

“Data” Category



Request data

The information to be displayed within the application is requested on the basis of the entered selection criteria. This process might take some time depending on the dataset from which the system filters data and on the selection result to be transferred and displayed.



Cancel

The query sent by clicking the “request data” button can be canceled using this function.

**Print preview**

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

**Save**

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

"Functions" Category**Add**

Adds a new article.

**Copy**

Copies the selected article.

**Edit**

Edits an already existing article

**Delete**

Deletes the selected or several selected articles.

"Help" Category**Help on operation**

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "[moc_cc.pdf](#)". It describes how to use MOC in general and applies for all applications.

**Help on application**

This function opens the manual corresponding to the respective application from which the help was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.



Help on detail application

This function opens the application manual at the section where the respective detailed application is described.

3.4 Selection parameters

The application provides the following selection criteria:

“General“ tab

- Article no.:
Article number
- Drawing issue number:
Drawing issue number of the article, often also referred to as index
- Designation:
Article name
- Inactive:
Inactive, active articles. The checkbox is not activated by default.
- Customer article no.:
Customer article number
- Article model:
Article model

“Groups“ tab

- Group:

The article group tree can be opened using the  button if an article group is to be filtered. There is a function to accept and cancel the activity.

3.5 “Article” Detail Application

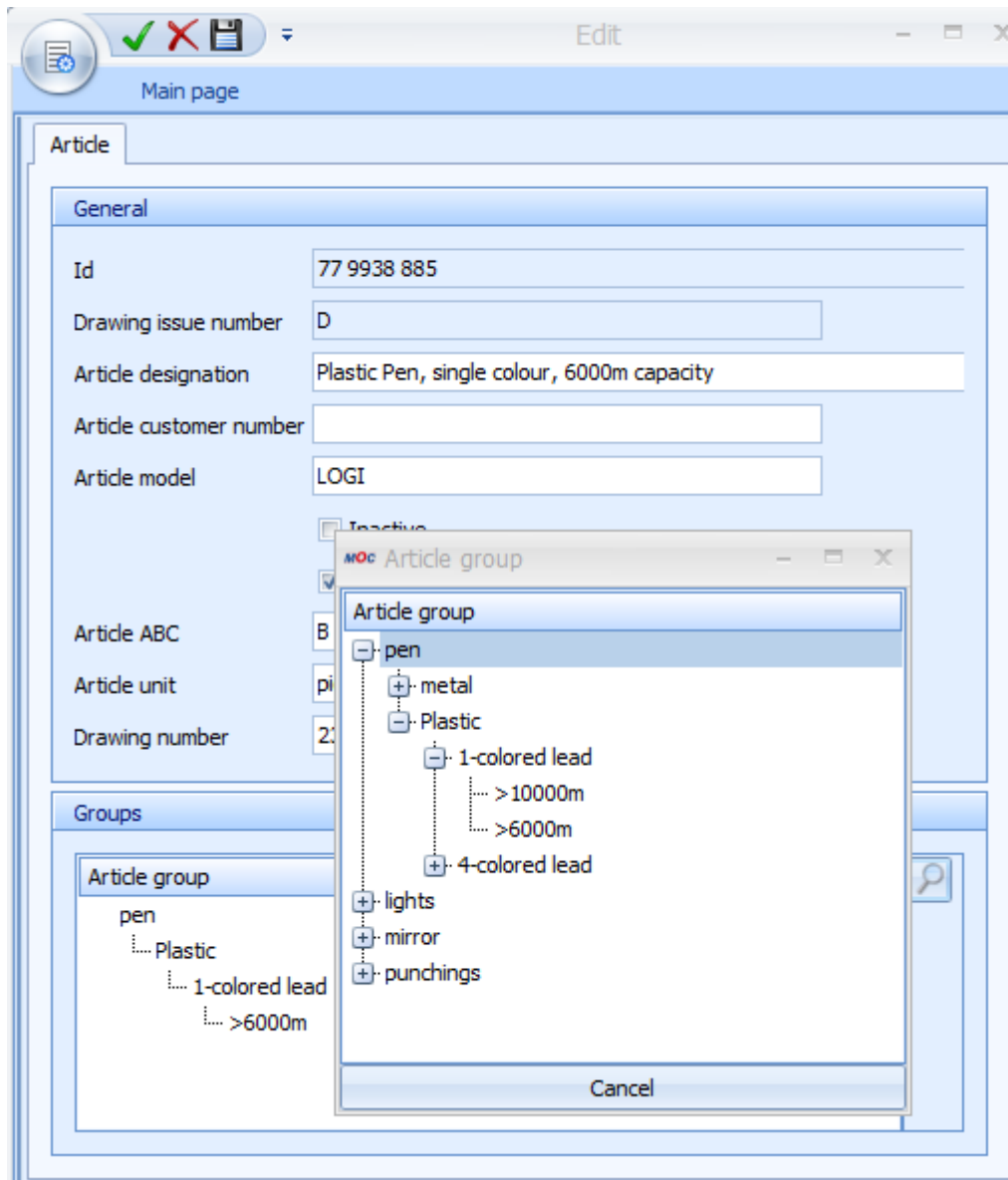
The article number as well as the drawing issue number uniquely identify articles in all areas of HYDRA-CAQ referring to the article catalog. The drawing issue number, also referred to as article index, may be very important, in particular, for inspection planning and when inspection orders are generated. Thus, it is possible to create, for example, an inspection plan for article 12938 with the drawing issue number A and B within inspection planning, whereas there are different inspection specifications for each drawing issue number. Unless the drawing issue number is indicated and thus may be part of the inspection plan, the system that generates the inspection requirements, must deliver this drawing issue number.

Consequently, the “article no.” and “drawing issue number” fields are key fields, i.e. when a new article is saved, it is first checked whether an article with this key information already exists.

By distinguishing between active and inactive articles, it may be defined whether or not the articles are available in certain selection lists. Thus, no inspection plan can be created for an inactive article. However, inactive articles may be evaluated at any time. Moreover, inactive articles can also be reactivated at any time.

Furthermore, an article can be defined as being subject to documentation. In addition the dialog provides the fields customer article number, article model, article ABC, drawing number as well as the possibility to assign units. The catalog of units is accessed to assign units (dimensions).

The corresponding group needs to be assigned, provided that article groups are to be evaluated or family inspection plans are to be used at a later point in time. Groups are assigned by opening the group tree using the magnifier button. Using the hierarchic tree entries the required group can be selected in the group tree and accepted by double clicking.



The assigned group including the hierarchical group structure then appears in the “groups” field of the editing dialog of articles.

When articles are displayed in a list, the group hierarchy is represented by the columns “group 1 to group 5”.

Groups are maintained in the “article groups” application and are described in the document entitled “[MOC_Groups.pdf](#)”.

4 Summary

4.1 General notes on the document

This document describes the “Groups“, e.g. article groups, application of the Manufacturing Operation Center (MOC). For general information on how to use MOC, please refer to the “[moc_cc.pdf](#)“ document.

5 Groups

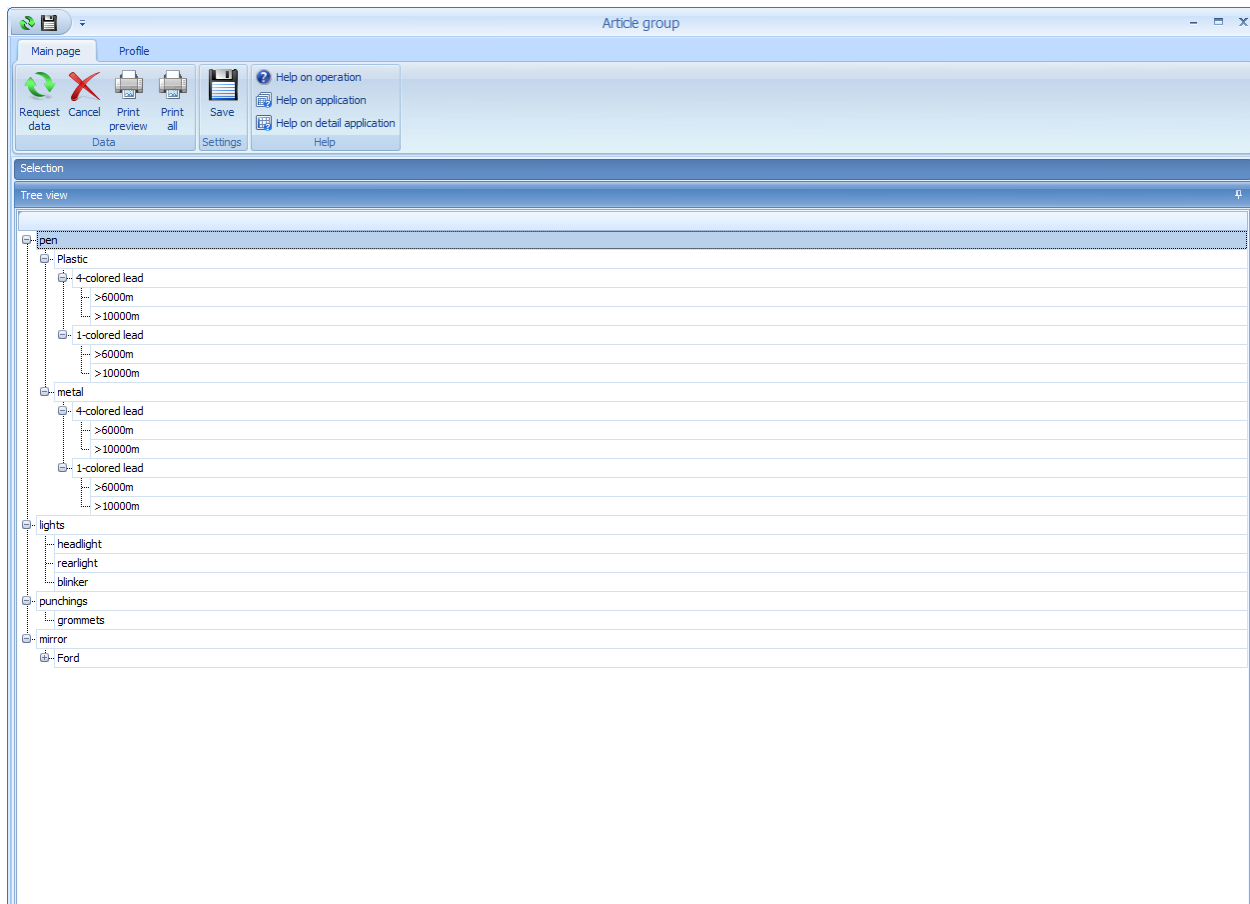
The group catalogs have been designed to create and edit groups for the different applications. The created groups may be assigned to master data of the corresponding application. Consequently, article groups may be created, for example, and assigned to the articles. In this case, it is also possible to create inspection plans on the basis of article groups.

Basically, the creation of groups is also reasonable for failure mode analyses.

5.1 Starting the function

Menu	Master data → Quality management → Article groups Master data → Process data processing → Article groups Master data → Quality management → Measure groups Master data → Quality management → Failure type groups Master data → Quality management → Failure location groups Master data → Quality management → Failure cause groups Master data → Quality management → Causer groups Master data → Quality management → Cost type groups
Transaction code	atcgr → Article groups measgr → Measure groups ftypgr → Failure type groups flocgr → Failure location groups fcaus → Failure cause groups origr → Causer groups costgr → Cost type groups
Function authorization	atcgr -> Article groups measgr.* → Measure groups ftypgr.* → Failure type groups flocgr.* → Failure location groups fcaugr.* → Failure cause groups origr.* → Causer groups costgr.* → Cost type groups

5.2 Default Application Layout



This figure of the article group catalog is exemplary for all groups.

5.3 Toolbar

The toolbar contains the function calls that are available for this application and possibly links to other applications. The functions placed on the “general” tab of the toolbar refer to all detail applications. In addition to the standard functions such as “help”, “request data”, “save application settings” and “print preview”, the other tabs include specific functions that are tailored to the corresponding detail application. The individual functions of the application are listed in the paragraphs that follow.

"Data" category



Request data

The information to be displayed within the application is requested according to the entered selection criteria. This process might take some time depending on the data set from which the system filters data and on the selection result to be transferred and displayed.

**Cancel**

This function cancels the query sent by clicking the "request data" button.

**Print****preview**

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

**Save**

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

"Functions" category

There are no special functions for this detail application. Groups are created, changed and deleted using the context menu of the right mouse button within the tree structure view.

"Help" category

**Help on operation**

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "moc-cc.pdf". It describes how to use MOC in general and applies for all applications.

**Help on application**

This function opens the manual that describes the application from which the help function was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.

**Help on detail application**

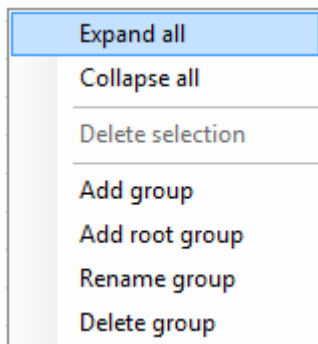
This function opens the application manual at the section where the respective detailed application is described.

5.4 Selection parameters

There are no selection parameters. A specific group can be found by using the “fast selection” function. To use the “fast selection” function just open the group tree structure on the 1st level, select the first entry and enter the first letter of the group in question. Consequently, the first group starting with this letter that is found is selected. The “fast selection” function also integrates subordinate groups that are not opened. If the requested term is included in a group that is not opened, it will be opened automatically.

5.5 "Groups" Detail Application

A group may be created, changed or deleted by opening the context menu of the right mouse button within the display area of the tree structure.



It is altogether possible to define groups up to the fifth hierarchy level. The “add root group” function has to be selected in the context menu of the group tree to create a new main group (1st level). The menu entry “add group” generates a sub-group (level 2 to 5). A designation, which is directly entered in the list view, has to be assigned for this new group. To be able to save the new group, click above or below this new entry within the group tree. Then a confirmation prompt appears asking whether or not the new group is to be saved. Provided that this question is affirmed (“yes”), the entry is saved. The same applies for renaming of groups. Regardless of which hierarchy level is concerned, the entry has to be selected to be able to edit the group designation. Changes can directly be entered in the corresponding line of the tree view. Click above or below the entry to be changed to be able to save the modification. No confirmation prompt appears when it comes to renaming.

An entry is also deleted by selecting a group entry and executing the “delete group” function. Only the group that is at the bottom of the group tree can be deleted.

The “expand all” context menu entry opens all groups up to the lowest hierarchy level. The “collapse all” context menu option closes all entries up to the first level.

The "delete selection" function cannot be used in the maintenance of groups dialog. This function is enabled, for example, in the maintenance of articles application if an article group is selected and this selection is to be removed/deleted.

6 Characteristic Master Data

Summary

Menu	Master Data → Quality Management → Characteristics
Transaction code	chrq
Function authorization	chrq

Utilization

The characteristics catalog has been designed to define characteristics and, as a result, to predefine characteristic data of inspection plans. For this reason, it is directed to people involved in inspection planning.

The characteristics catalog is one of the most important basic catalogs for without this catalog no inspection plan could be created. As it has been designed to predefine characteristics data for inspection plans it provides extensive input possibilities. Generally, only this data, which has not to be modified at a later point in time when assigning it in the inspection plan, should be entered in the characteristics catalog. The definition of limit values, for example, is usually not reasonable as they are only known when the inspection plan is created. For a relation to a specific article is only established with the assignment to an inspection plan. If you keep this always in mind it becomes obvious which information needs to be defined. For this reason, it should be considered thoroughly whether the characteristic "outer diameter" is to be created only once and the detailed information is to be defined later in inspection planning or whether it would be better to create several characteristics for the "outer diameter" specifying the limit values already. In the majority of cases, it is better to only enter a few general characteristics. The required evaluations/reports also play a role in this. If a new "outer diameter characteristic" is created for nearly every tolerance modification it is almost only "applicable" for one article and comprehensive evaluations are not possible when failures are analyzed later"

It is important that each detail defined here can be modified in the inspection plan or that details, which have not been stated, can be added.

The configurations made in the characteristic master data are not final. Characteristic master data has rather been designed as a template for inspection planning. All settings of the characteristic master data can be supplemented and modified in inspection planning.

Integration

The characteristic catalog is a global catalog that is used in many QM applications. Please find below some possible fields of application that refer to the characteristic catalog.

- Inspection planning for production, goods receipt, goods issue, initial samples and calibration
- Inspection requirements for production, goods receipt, goods issue, initial samples and calibration
- Failure analysis in complaints management
- Several reports/evaluations

Prerequisite

There are no special requirements.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Characteristics number
- Characteristic designation:
Designation of the characteristic – Please note: wildcards may be used "*"
- Characteristic type:
Inspection type: attributive, inspection chart, variable

"Details" tab:

- Gages

Selects a gage
- Gage designation:
Selects a gage designation

"User fields" tab:

- If user fields are created they may be selected

If several selection criteria are used overlapping results are displayed in the characteristic master data.

In addition, the column filter allows for the content of each individual column to be filtered.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

"Characteristics" tab

Characteristic no.

Unique number of the characteristic

Characteristic designation

Designation of the characteristic

Input type

Automatic or manual collection type. This field controls the release of HYDRA-PDV fields (for automatic collection). If the automatic collection function is selected, the characteristic type is restricted to the "variable" option.

Characteristic type

Decides whether the inspection is based on the collection of measured values (variable) or on the indication of the number of identified errors (attributive). When it comes to the attributive inspection, the decision is often based on "pass" or "fail". The inspection chart and the information characteristic are further characteristic types. The information characteristic has only been designed to display a document while the inspection is running. Subject to the input type, the lower area of the dialog provides corresponding sampling schemes.

Inspection result base

Decides whether all samples or only the last sample are used to determine the inspection result (pass/fail).

Mandatory inspection

If this checkbox is activated, at least one measured value has to be entered for this characteristic, before an inspection order can be completed with this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

The authorization "iriscp.formula" is required to show this field in inspection plan characteristics.

Formula:

Please also see the section entitled "formula calculation"

The authorization "iriscp.formula" is required to show this field in inspection plan characteristics.

"Details" tab**"Gage" group****Gage**

Defines whether a gage or gage group is to be assigned to the characteristic:

Assignment of the gage to be used (or even gage group).

The resource management may be accessed using the resource type "PRM". The authorization "iriscp.gage" is required to show this field in inspection plan characteristics.

Gage designation

Shows the gage designation

"Properties" group**Certificate printing**

The corresponding definition determines whether this characteristic is to be printed (show selection or print always) or not (print never) when certificates are printed out at a later stage (e.g. acceptance, inspection certificate). If the "display selection" option is chosen, a list of the characteristics that are assigned to this flag is displayed prior to printing. These characteristics have been predefined for printing in the list. However, this selection may be removed. Finally, all selected characteristics and the characteristics assigned to the "print always" option are printed out on the certificate. Characteristics assigned to the "print always" flag do not appear in a selected list, as they are printed in any case. However, it is important to note that these certificate flags only affect certificate forms.

Error weighting

For information purposes, a rating can be made here if the inspection result for this characteristic is "fail".

"Inspect" group**Analysis selection catalog**

Using this selection function, an analysis selection catalog may be selected. This catalog restricts the selection options when failures are entered (failure types, failure locations, etc.). (But nevertheless, all available failures may be entered by directly entering their number).

Designation of analysis selection

Shows the designations of analysis selection catalogs

"Specifications" tab

Once the "specifications" tab has been selected, the sample scheme and dimensions may be entered. In this context, it has to be considered that (as already mentioned) the definition of tolerance limits in the master data of characteristics is only reasonable if certain conditions are met. The same applies to the definition or calculation of action and warning limits. The different possibilities are described anyway.

"Sampling scheme" group**Sampling scheme**

The following sampling schemes are available:

- 100% inspection
- k value inspection
- Batch inspection
- n-c inspection
- SPC inspection

The sampling scheme determines the course of the inspection. With an n-c inspection with the parameters 5-0, 5 items are checked and 0 failures may be detected.

The section entitled "Sampling scheme" describes this process in more detail.

Sample size/assumable sample size

Specification of the sample size or assumable sample size depending on the sampling scheme, see section "sampling schemes".

Acceptance quantity

Acceptance quantity for the n-c inspection, please also see section "sampling schemes".

Interval type

Input for SPC or n-c inspections: time, pieces, once, none. Please also see the section "sampling schemes".

Interval value

Specifies the interval subject to the interval unit.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

Interval unit

For n-c or SPC inspections, e.g. minutes, hours.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

With output batch change

Causes an inspection to become due when output batches change.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

With machine status change

Causes an inspection to become due when machines change.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

Source status

Specifies the source statuses, separated by commas, (specific, non-productive machine statuses), which are to cause an inspection to become due when switching into a productive machine status.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

With the change of shifts

Causes an inspection to become due when shifts change.

The authorization "iriscp.interval" is required to show this field in inspection plan characteristics.

"Construction measures" group**Unit**

Pieces, meter, kg, etc. Unit of the characteristic. Units are assigned by accessing the unit catalog.

Decimal places

Number of decimal places. Leading zeros in front of the comma are not displayed in the specification fields. The number of decimal places that is defined within system settings is indicated by default.

Size

Plausibility and tolerance limits can be entered as absolute, relative or percentage values. Please note that relative or percentage lower limits (lower tolerance limit, lower process limits) have to be indicated with a negative algebraic sign.

Norm

Computation of tolerances on the basis of specific norms (e.g. ISO metric fits). Subject to the selected norm, further information is requested (e.g. engineering fit). Tolerance limits are computed automatically on the basis of these specifications.

Engineering fit

Computation of tolerance limits on the basis of a specific norm and fit. The selected fit depends on the selected norm.

UPL

Specifies the upper plausibility limit

UTL

Specifies the upper tolerance limit (upper specification limit)

Target value

Indicates the target value

LTL

Specifies the lower tolerance limit (lower specification limit)

LPL

Defines the lower plausibility limit

Generate error (UTL)/(LTL)

If the checkbox "generate error" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:TG> or AUTO:TG<) to be defined when measured values are recorded. This option is not available for attributive characteristics, as the specification is only used for information purposes in this case.

"User fields" tab

In case user fields are defined for characteristics, they are displayed and may be edited here.

"Chart 1/chart 2" tab

The control charts to be used can be defined in the "chart 1/chart 2" tab. These control charts are later available in the measurement recording for terminals (AIP). Altogether two different control charts may be defined, whereas the action limits, warning limits and the mean value may be defined for each control chart if variable characteristics are in use. There are two different possibilities to define these limit values. They are either entered manually or determined automatically on the basis of specific default values that are defined in the "default values" tab "chart 1/2". For further information on control charts, please see the sections entitled "control chart for variable characteristics" and "control charts for attributive characteristics".

Chart 1 / chart 2

Defines the control chart that is to be displayed in the measurement recording dialog at the terminal. Action limits may also be defined on the basis of the control chart type.

Upper AL

Specifies the upper action limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Upper WL

Defines the upper warning limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

MV (Mean value)

Specifies a mean value, e.g. as basis for the automatic calculation of limits by the system.

Lower WL

Defines the lower warning limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Lower AL

Specifies the lower action limit. It can be determined by the system on the basis of default values, provided that the "calculate" checkbox is enabled. (Please also see section "control charts for variable characteristics").

Generate trend error

The option "generate trend error" has to be activated to be able to generate an automatic error if a trend exists (e.g. seven values in a row are descending or ascending, the number of values is defined while the system is customized). To determine a trend, the samples of an inspection step characteristic are taken into account, sorted by the sample number, regardless on which machine data has been recorded.

Generate error (UWL) / (LWL)

If the checkbox "generate error (UWL) / (LWL)" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:WG> or AUTO:WG<) to be defined when measured values are recorded. A violation of the limit value refers in this case to the defined control chart. If, for example, an xq chart is defined an automatic error will only be generated, provided that the corresponding xq value of the control chart, not the single value, is beyond the warning limits.

Generate error (UAL) / (LAL)

If the checkbox "generate error (UAL) / (LAL)" is enabled it is defined whether a violated limit value automatically (in the background) results in the failure type "limit value violation" (AUTO:EG> or AUTO:EG<) to be defined when measured values are recorded. A violation of the limit value refers in this case to the defined control chart. If, for example, an xq chart is defined an automatic error will only be generated, provided that the corresponding xq value of the control chart, not the single value, is beyond the action limits.

"Default values chart 1 / default values chart 2" tab

For further information on control charts, please see the sections entitled "control charts for variable characteristics" and "control charts" for attributive characteristics.

"Default for calculating limit values" tab**Calculation type**

Default values to compute limit values: Cpk, Sigma, sQuer/an, RQuer/dn, relative deviation of XQuer, percentage deviation of XQuer

Cpk

Default value of cpk

Sigma

Default value or computed sigma value

RQuer /sQuer

Default value for RQuer/sQuer

"Non-action probability" group**Action limits (non-action probability)**

Selects the action probability (only visible with the computation type: cpk, Sigma, RQuer /sQuer)

Warning limits (non-action probability)

Selects the action probability (only visible with the computation type: cpk, Sigma, RQuer /sQuer)

"Deviation from XQuer specification" group**rel. AL**

Direct entry of the action limits (only visible if the calculation types "rel. deviation/deviation in percent" are used)

rel. WL

Direct entry of the warning limits (only visible if the calculation types "rel. deviation/deviation in percent" are used).

"Confidence interval" group**Confidence interval**

One-sided or two-sided. It may be chosen between "on-sided" and "two-sided" for the control charts R and s.

"XQuer" group**XQ**

Target value, tolerance center, mean value of xq chart, input (only visible and can only be selected with an xq control chart)

Editing functions

The below screenshot is one example for how an editing dialog could look like. The design and alignment of fields may deviate from that of this screenshot.

The screenshot shows the 'Edit' dialog box in the Complaint Management software. The dialog has a title bar with a green checkmark, a red X, and a dropdown arrow. Below the title bar is a 'Main page' tab. The dialog is divided into two main sections: 'Sampling scheme' and 'Constructional measures'.

Sampling scheme

- Sampling scheme: n-c-inspection
- Sample size: 5
- Acceptance qty.: 0
- Interval type: ☐ once ☐ none ☐ piece related ☒ time related
- Interval value:
- Interval unit: Minute(s)
- ☐ With output batch change
- ☐ With machine status change
- ☐ With change of shifts

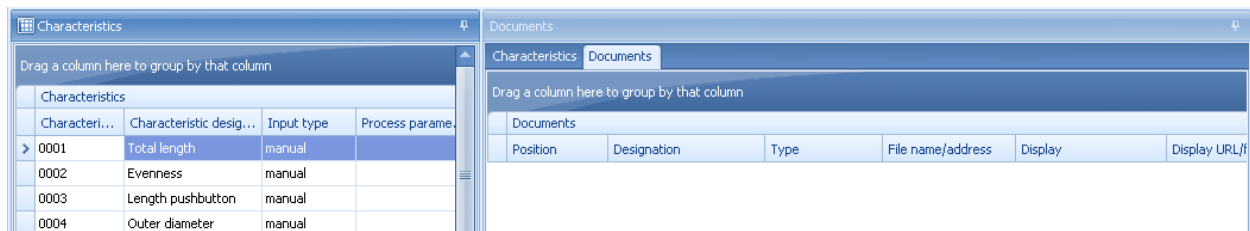
Constructional measures

- Unit: mm
- Decimal places: 5
- Size: ☒ absolute ☐ percentage ☐ relative
- Norm:
- Upper PL: 135.0
- Upper TL: 130.25
- Target value: 130.0
- Lower TL: 129.9
- Lower PL: 125.0
- ☒ Generate error (UTL)
- ☒ Generate error (LTL)

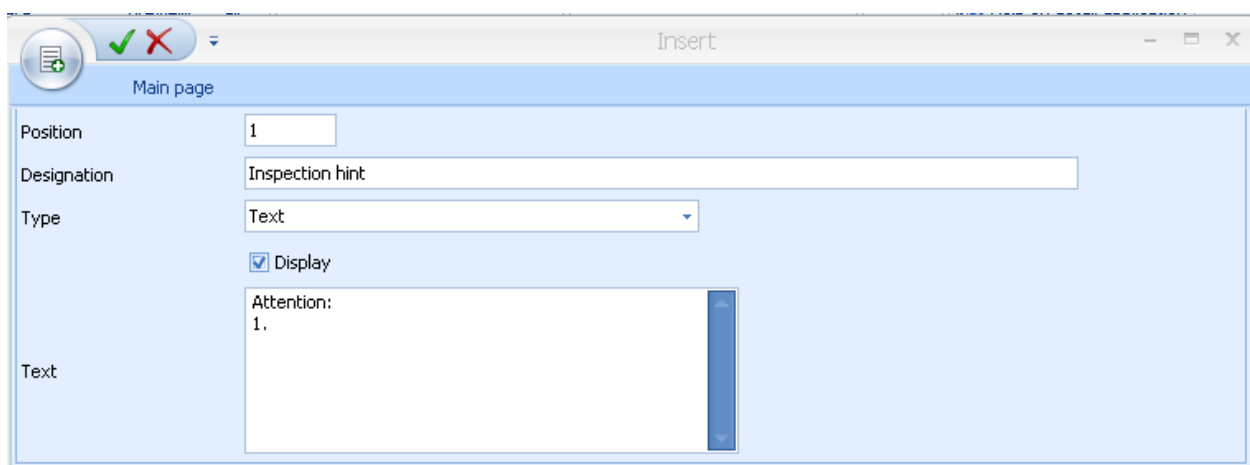
Toolbar

There are no further specific function keys in addition to the standard features.

"Documents" detail application



Provided that the "documents" tab has been activated, as many documents as required may be assigned to each characteristic. If this tab is activated corresponding editing buttons are enabled in the toolbar to edit the documents.



When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

"URL" and "text" are provided as file types. The file name including the path may be entered for the "file" type. The file type "URL" allows for the Internet or intranet to be accessed. The third file type "text" allows for the text to be entered directly.

Please note:

The different variants of the file type "URL" supported by the shop floor client are described in the relevant manual dealing with the shop floor client. It might, for example, be the case that "https" URL entries can be displayed on MOC but not on the AIP shop floor client.

A designation may be assigned to each defined document. Moreover, it can be determined in which order the documents are to be listed. This is made by the "position" field (numeric input). The specifications have to be unique within this list. In addition, the "display with inspection" checkbox determines whether or not the document can be displayed during the inspection process.

In this case as well it has to be reflected thoroughly whether documents are to be assigned without a precise reference to an article. Normally, document assignment depends on the article.

“Documents“ toolbar

In addition to the standard features, the “Documents” detail application also provides a button to view the documents.



Show documents

With this button it is possible to open and show a document, which is attached by a link. In this case, however, a program needs to be installed that is able to open the linked file type, e.g. Acrobat Reader for PDF documents.

6.1 Sampling scheme

The user can choose from five sampling schemes in a specified list. Subject to the selected sampling scheme, some additional information has to be defined. Depending on how they are used in the different inspection plan areas (e.g. production, goods receipt, goods issue) only a subset of these sampling schemes is available.

Sampling scheme n-c inspection: The sample size is entered in the "sample size" field (= n) and the maximum number of admissible non-conforming units is entered in the "acceptance quantity" (=c) field. The figure "c" is defined as acceptance number. This means: if n = 50 und c = 1 the characteristic, and thus the item, is only rated as "fail" if two non-confirming units occur, provided that the sample size amounts to 50.

Sampling scheme 100% inspection: The sampling scheme "100% inspection" is generally only used in the goods receipt and goods issue. In this case, the sample size is computed from the actual quantity of the inspection requirement and corresponds to it.

Sampling scheme SPC inspection: The sampling scheme "SPC inspection" nearly corresponds to the "n-c" inspection plan. The only difference is that the acceptance limit "c" is not used in this case.

Sampling scheme batch inspection: The sampling scheme "batch inspection" only applies to the areas "goods receipt" and "goods issue" in the standard configuration. The percentage specifying how much percent of the batch is to be checked is entered here. Later in the inspection order characteristic the sample size is calculated from the actual quantity of the inspection requirement and multiplied by the specified percentage.

Provided that action limits are to be calculated, the expected (assumable) sample size has also to be entered here.

Sampling scheme k-value inspection: With the k value inspection the entered k value is checked against the computed k value and if this value is violated the sample is rated "fail".

6.2 Control charts for variable characteristics

The variable characteristics provide the charts $\bar{x}$, s and R.

The production dispersion is used in many computations of the statistical quality assurance. One example is the calculation of capability indices and action limits of a quality control chart. Vice versa, it is possible to estimate the dispersion of production, provided that a process capability index is indicated, and the action limits are calculated on this basis.

The specifications for the calculation of limit values can be found in the tab "default values chart 1" or "default values chart 2", where values to estimate the production dispersion can be entered. The action and warning limits can be calculated on the basis of these specifications. However, it is also possible to enter the production dispersion directly. This program provides altogether three computation options.

At first, the specifications are described on the basis of the $\bar{x}$ and s chart. The differences with the R chart are explained in more detail in the sections that follow.

There is often a specification for the process capability index cpk. This specification is reasonable, as respecting the process capability index cpk implies that items can be produced within the range of tolerance. On the basis of the default cpk value, the system computes internally an estimated value for Sigma, which is entered to the right of the "Sigma" option for information purposes. This computed basic estimated value has in turn been designed to calculate the limit values of the $\bar{x}$ /s chart. The computation is performed, once further data has been entered using the "calculate" button. The calculation method "cpk" is set by default. In addition, there are also the calculation methods "sigma" and sQuer/an".

This cpk value of 1,33 ensures that 99.725% of the characteristic values are within the tolerance. However, it also often required that 99.994% of the characteristic values are within the tolerance limit, which corresponds to a cpk value of 1.67.

The calculation method sq/a_n means that an estimate of the standard deviation is calculated from the quotient of the medium standard deviation and a correction factor a_n . This correction factor depends on the sample size, which is expressed by the index n . The values for a_n are defined in the system and are requested automatically. This estimate of the standard deviation is best in case that there is no specification of the process capability index and the production dispersion is unknown and thus the specification of the sq -value has still to be corrected by a correction factor. Moreover, it is the most efficient one under the given conditions. The sq -value has also to be specified for a subsequent calculation of limit values. This can be entered in the field on the right next to the sq/a_n option. If the "calculate" button is clicked in the preceding dialog the estimate sq/a_n is calculated from the specification of sq and it is entered on the right next to the sigma option for information. This estimate is then the basis for the calculation of action and warning limits of the xq/s chart

The third calculation method requires a sigma value being specified. In this case it is assumed that sigma is known and consequently the correction factor is not required. To the right of the "sigma" option the sigma value is entered. In contrast to the previous method sq/a_n is replaced by sigma. In cases of doubt and since in the majority of cases sigma is not really known, the calculation method based on the specification of sq including the automatic determination of the estimate sq/a_n should be performed for variances.

If the "relative deviation from Xq " or the "deviation from Xq in percent" is selected as "specification for calculating limit values" the input option for "action probability in %" disappears and the "deviation from target value" can be entered instead. Limit values are then calculated on the basis of these values and the default calculation of Xq (target value, tolerance center, mean value of XQ chart, input).

Further details have to be made in order to determine action and warning limits of the Xq chart. An Xq value has to be specified. The system offers the possibilities to equate the Xq value with the tolerance center or the target value or to specify a value manually. If the process is supposed to be aligned to the mean value the tolerance center should be preferred as Xq value.

The action probability must be entered in per cent in order to calculate action and warning limits of the Xq/S -chart. For this purpose, it has to be defined beforehand whether the calculation is to be made on the basis of one-sided or two-sided limit values. One of the two options has to be selected.

Having defined the "one-sided" or "two-sided" option, the action probability has to be entered in per cent. All possible and reasonable action probabilities are defined in the system and need only to be selected from the list. The computation of the Xq chart is based on normal distribution. The value 99,725 is to be selected if, e.g., 99,725% of characteristic values are supposed to lie within the action limits. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed along with the respective action probability for information.

If warning limits are also to be calculated an action probability has also to be entered here. Please take into account that the probability value of the warning limit has to be less than the action limit value.

No sigma area is displayed in the selection list since the distribution of χ^2 is considered for the calculation of limit values of the s-chart. Apart from that, the entry is the same as for the Xq chart.

The calculation of limits is triggered as soon as the specifications made are saved, provided that the "calculate" checkbox has been clicked before.

As already mentioned, the user is also able to enter the limit values directly without any defaults.

If the R-chart is selected instead of the Xq or s-chart, the option sq/a_n is replaced by Rq/d_n in the specifications for the calculation. The calculation of the estimate of sigma is made on the basis of the default mean range R divided by the correction factor d_n . This correction factor in turn depends on the sample size n. The corresponding values are defined in the system and are selected automatically. Apart from that, the rest is the same as for the Xq or s-chart. Please note that limit values are not calculated on the basis of a χ^2 distribution for the R-chart but on the basis of a table defined in the system, which is based on standardized ranges.

6.3 Control charts for attributive characteristics

The p- and u-charts are available for attributive characteristics.

p designates the share of non-conforming units in the sample and u designates the failures per unit in the sample. It is important for the p-chart that each item is either described as error-free or faulty. If an item shows several failures it is referred to as faulty only once.

In contrast to variable characteristics there are no lower limit values. Furthermore, it is normally not necessary to state the values UTL, LTL and target value.

A pq or uq value has to be entered in per cent to calculate specifications automatically. This can be done in the default values dialog.

The calculation of limits is triggered as soon as the specifications made are saved, provided that the "calculate" checkbox has been clicked before.

Calculation is respectively based on normal distribution. The value 99,725 has to be selected if, e.g. 99,725% of the characteristic values are supposed to lie below the upper action limit. As the specification of a sigma area is commonly used by some users the corresponding sigma area is displayed with the respective action probability for information.

6.4 Computation of formulas

On the basis of an entered formula, measured values can be calculated automatically by way of measured values or statistic values of other characteristics that have been checked before.

The first part of the formula specifies where the calculation of formulas is supposed to take place. The following types are available:

- V – Calculation on the level of single values (Value).
A single value is generated for the calculated characteristic for each single value of the characteristics involved.
- S - Calculation on the level of samples (Sample).
Exactly one single value is generated for the calculated characteristic for each sample of the characteristics involved.
- C - Calculation on the level of characteristics (Criteria).
Exactly one single value is generated for the calculated characteristic (with respect to the overall statistic of all characteristics involved)

The actual formula follows this identifier. The following operators, functions and constants are supported in this context:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates natural logarithms
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates common logarithms
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
$x + y$	Addition
$x - y$	Subtraction
x / y	Division
$x * y$	Multiplication
$x ** y$	Calculates x raised to the power of y
Absolute term (constant)	
pi	3.141592654
e	2.718281828

If constant, numeric values are used in formulas it has to be taken into account that no separators are used for thousands digits. If these constants are floating-point numbers it has to be considered that they use a dot as decimal separator instead of a comma.

The following syntax applies for the variables identifying the single values or statistic values of the order characteristics involved [x:y:z]. The x parameter identifies the statistic value to be used. The available values are listed as follows. Please consider possible restrictions regarding the calculation level.

- **X** – Single value
(is only available for calculations on the level of single values)
- **AVG** – Mean value
(is only available for calculations on the level of samples or characteristics)
- **MIN** – Minimum
(is only available for calculations on the level of samples or characteristics)
- **MAX** – Maximum
(is only available for calculations on the level of samples or characteristics)
- **SUMX** – Sum of single values
(is only available for calculations on the level of samples or characteristics)
- **R** – Range
(is only available for calculations on the level of samples or characteristics)
- **S** – Standard deviation
(is only available for calculations on the level of samples or characteristics)
- **N** – Sample size
(is only available for calculations on the level of samples or characteristics)
- **M** – Number of samples
(is only available for calculations on the level of characteristics)

The y parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- **SENO** – identification via the MS (maintenance sequence/operation sequence) of the characteristic (serial number)

- INCR – Identification via the characteristic number (inspection criteria)

If the characteristic number is not unique within the inspection requirement it is not predictable which one of the applicable characteristics is used at the time of calculation.

The parameter z identifies the characteristic on the basis of the field content determined by parameter y . Either the MS/OP number or the characteristic number of the calculation source is written in this field. If the characteristic number has a blank it should be replaced by an underscore within the formula.

Example 1:

A new characteristic is computed from the single values of the characteristic assigned to the number "LENGTH"/"LAENGE" divided by 2.5. A corresponding single value is supposed to be calculated for each single value of the source characteristic (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] / 2.5

Example 2:

The characteristic "surface" results from the product of the characteristics with the characteristic number "LENGTH"/"LAENGE" and "WIDTH TOTAL"/"BREITE GES". A single value of the characteristic "surface" is supposed to be calculated for each single value of both source characteristics (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] * [X:INCR:BREITE_GES]

Example 3:

The characteristic "maximum margin width" results from the subtraction of the minimum of the characteristic "inside diameter" (MS 10) from the maximum of the characteristic "outside diameter" (MS 20). A single value of the characteristic "maximum margin width" is supposed to be calculated for each sample of both source characteristics (calculation on the basis of samples).

→ Formula: S: [MAX:SENO:20] - [MIN:SENO:10]

The escalation CPAUMW.CALCULATED_CRITERIAS_GET_VARIABLE_VALUE is triggered if unknown variables are used within a formula (faulty parameters x and/or y). However, the escalation management module has to be licensed in this case.

When it comes to the calculation of formulas, it is allowed to compute new calculated characteristics from calculated characteristics that have already been computed. However, this nesting may not have more than 10 references one below the other. Furthermore, double concatenations must not be created (Example: characteristic A is calculated from characteristic B and characteristic C; characteristic C is calculated from characteristic A).

Tool numbers, machine numbers or cavity numbers, etc. are not defined for the single values computed.

The same number needs to be assigned to all source samples of the calculation to take over a corresponding number (batch number, sample number, serial number, etc.). If there is no number that is assigned to all source samples a number will not be assigned to the calculated sample. Provided that several numbers are found that are assigned to all source samples only the first number that has been found is assigned to the calculated sample. This function only applies to numbers, which have been assigned on sample level.

7 Customers

Summary

Menu	Master data → Quality management → Customers
Transaction code	cto
Function authorization	cto

The customer catalog has been designed to edit/keep customers. Provided that there is an interface to a higher-level system (e.g. ERP system), customers can be created automatically via interface. As soon as a new customer is created or changed in the ERP system, for example, the customer data record is automatically created or changed in the customer catalog including the defined information.

Utilization

The customer number uniquely identifies customers in all QM applications that refer to the customer catalog. The customer catalog is used as basis in particular for inspection requirements of production and and the goods issue as well as for the complaint management.

The “customer number” field is the key field, i.e. if a new customer is saved it is verified whether a customer with this key information exists already.

By distinguishing between active and inactive customers, it may be defined whether or not they are available in certain selection lists. Thus, no complaint can be created for an inactive customer, for example. However, inactive customers may be evaluated at any time. Moreover, inactive customers can be reactivated at any time.

Extensive address and contact data can be defined in addition to the customer number and the designation.

If a customer is designated as “party in charge” they will be included in the selection list for the parties in charge. Such selection lists are integrated in different detail applications. The list of the parties in charge is accessed mainly in the complaint management function and when measures are generated.

Integration

Customer data is a global catalog that is used in many QM applications. The below list shows the applications referring to the customer catalog.

- External people
- Departments

- Production inspection planning
- Production inspection requirements
- Complaint management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2, and 3 are available in addition to the customer number and customer name.

The active or inactive customers can be restricted using the filter field "inactive".

Field Descriptions

The available fields are self-explanatory and are not described separately, except for the address fields.

The content of the individual address fields is not specified by default and, as a result, may be defined by the user. Normally, address field 1 should include further details on the company, e.g. "site X". As there is no field defined for the street, address field 2 or 3 (to be preferred) is to be used for the street including street number.

Editing functions

The below dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a 'Main page' tab. The content is organized into two sections: 'Company' and 'Administration'. The 'Company' section contains the following fields:

- Type: Customer (dropdown menu)
- Company number: 893239
- Company name: Office GmbH & Co. KG
- Address 1: Lor 11, #01-235
- Address 2: (empty text box)
- Address 3: (empty text box)
- ☐ Inactive
- ☒ Party in charge
- P.O. box: (empty text box)
- ZIP code: 57665
- City: Example City 1
- Federal state: Johor
- Country: Malaysia
- Phone: +65 6475 9678
- Fax: +66 2 3456 7123
- Cell phone: (empty text box)
- Pager: (empty text box)
- E-mail: office@sample1.com.sg
- Web addr.: www.sample4711.com.sg

Toolbar

There are no other special function buttons in addition to the standard functions.

8 Manufacturer

Summary

Menu	Master data → Quality management → Manufacturer
Transaction code	mft
Function authorization	mft

The catalog of manufacturers has been designed to edit and update the list of manufacturers. Provided that there is an interface to a higher level system (e.g. ERP system) the manufacturers can be created automatically by interface. As soon as a new manufacturer is created or an existing manufacturer is changed in the ERP system, the corresponding manufacturer record is automatically created or changed with the defined information in the catalog of manufacturers.

Utilization

The manufacturer number uniquely identifies the manufacturers in all QM applications that refer to the catalog of manufacturers. In particular, the inspection requests of the goods receipt refer to the catalog of manufacturers.

The "manufacturer number" field is the key field, i.e. while saving a new manufacturer, the system checks whether there is already a manufacturer with this key information.

By differentiating between active and inactive manufacturers, it can be defined whether or not they are still to be available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive manufacturer. However, it is possible to evaluate inactive manufacturers at any time. Moreover, inactive manufacturers may be reactivated at any time.

Extensive address and contact data may be defined in addition to the manufacturer number and designation.

If the manufacturer is designated as being a "party in charge" by clicking the corresponding field, the manufacturer is included in the selection list for the parties responsible. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Manufacturer data is a global catalog that is used in many QM applications. Please find below some applications that refer to the manufacturer catalog.

- External people

- Departments
- Goods receipt inspection planning
- Goods receipt inspection requests
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the manufacturer number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive manufacturers.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there's a 'Main page' header. Below it, there are two tabs: 'Company' and 'Administration', with 'Administration' currently selected. The form contains the following fields and controls:

- Type:** A dropdown menu showing 'Manufacturer'.
- Company number:** A text box containing '827363'.
- Company name:** A text box containing 'Spanner International Pte Ltd'.
- Address 1:** A text box containing '5 Ayer Rajah Crescent'.
- Address 2:** An empty text box.
- Address 3:** An empty text box.
- Inactive:** An unchecked checkbox.
- Party in charge:** A checked checkbox.
- P.O. box:** An empty text box.
- ZIP code:** A text box containing '62783'.
- City:** A text box containing 'Example City 2'.
- Federal state:** A text box containing 'Sarawak'.
- Country:** A text box containing 'Malaysia'.
- Phone:** A text box containing '+65 6341 8679'.
- Fax:** A text box containing '+66 2 6574 9900'.
- Cell phone:** An empty text box.
- Pager:** An empty text box.
- E-mail:** A text box containing 'info@sample45.com.sg'.
- Web addr.:** A text box containing 'www.sample4712.com.sg'.

Toolbar

There are no other special function buttons in addition to the standard functions.

9 Suppliers

Summary

Menu	Master data → Quality management → Suppliers
Transaction code	sup
Function authorization	sup

The supplier catalog has been designed to edit and update the list of suppliers. Provided that there is an interface to a higher-level system (e.g. ERP system), suppliers may be created automatically by the interface. As soon as a new supplier is created or an existing supplier is changed in the ERP system, the data record for this supplier is automatically created or changed including the specified information within the HYDRA-CAQ supplier catalog.

Utilization

The supplier number uniquely identifies suppliers in all QM applications that access the supplier catalog. The supplier catalog is used, in particular, for inspection requests of goods receipt and the complaint management.

The "supplier number" field is the key field, i.e. while saving a new supplier, the system checks whether or not there is already a supplier with this key information.

By distinguishing between active and inactive suppliers, it may be defined whether or not the suppliers are available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive supplier. However, inactive suppliers may be evaluated at any time. Moreover, inactive suppliers may also be reactivated at any time.

In addition to the supplier number and designation, it is possible to specify comprehensive address and contact details.

Being identified as "party in charge", this supplier is included in the selection list for responsible parties. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Supplier data is a global catalog that is used in many QM applications. Please find below some applications that refer to the supplier catalog.

- External people
- Departments

- Goods receipt inspection planning
- Goods receipt inspection requests
- Complaints management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the supplier number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive suppliers.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified explicitly and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there's a 'Main page' header. Below it, there are two tabs: 'Company' (selected) and 'Administration'. The 'Company' tab contains a form with the following fields and values:

Type	Supplier
Company number	928762
Company name	SANYO Co. Ltd.
Address 1	Siglap Drive, #01-367
Address 2	
Address 3	
☐ Inactive	
☒ Party in charge	
P.O. box	
ZIP code	74663
City	Example City 5
Federal state	Sarawak
Country	Malaysia
Phone	+62 21 4659 6789
Fax	+60 7 6576 3311
Cell phone	
Pager	
E-mail	info@allied.com
Web addr.	www.theka.com.sg
Audit date	
Audit setting	
Audit value	

Toolbar

There are no other special function buttons in addition to the standard functions.

10 Distributor

Overview

Menu	Master data → Quality management → Distributor
Transaction code	distrib
Function authorization	distrib

The distributor catalog allows for different distributors to be defined and different master data to be assigned or referenced, e.g. external persons.

Utilization

The field content "distributor" enables unique identification.

The data record of a distributor catalog only includes a number and designation as well as a flag to identify it as "responsible". The possibility to assign different distributor entries is a decisive factor. The following entries of other master data may be assigned to a distributor.

- External persons
- Departments
- Customers
- Suppliers
- Manufacturers and
- Internal companies.

Once identified as being "responsible", the distributor is taken over into the selection list of the parties responsible. This list is integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures. The complaint management also allows for a team to be assigned. The content of this distributor catalog is provided in a selection list.

Integration

The distributor catalog is available in applications that enable assignment of responsible parties. Subject to the corresponding application or system configuration, it may be the case that "responsible distributors" are intentionally not included in the selection list.

Prerequisite

The master data that can be assigned to a distribution list need to be maintained before this function can be used in a reasonable manner.

Selection criteria

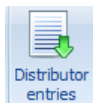
Selection criteria are self-explanatory and not described separately.

Field descriptions

The available fields are self-explanatory and not explained separately.

Toolbar

Distributor entries



Assignment of distributor entries

Function to assign distributors to a previously defined and selected catalog entry of the distribution list.

Detail application "distributor entries"

The application of distributor entries includes a list showing the data records that have already been assigned. The information on the selected data record is also shown in a display window.

Field descriptions

Name 1, Name 2, Name 3

The contents of the fields "name 1", "name 2" and "name 3" of the assigned distributor entry are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

All other fields are self-explanatory and not explained separately.

11 Failure Types

Summary

Menu	Master data → Quality management → Failure types
Transaction code	ftyp
Function authorization	ftyp

The catalog of failure types has been designed to describe the occurred deviations in more detail, e.g. deviations from specified limit values. In addition to the other failure catalogs (failure location, failure cause, causer), the failure type catalog is very important as it also includes inactive failure types by default. These inactive failure types are required for the generation of automatic failure types, e.g. in case a limit value has not been respected. For this reason, these inactive failure types must not be deleted. All inactive failure types that are important for the automatic generation of failures start with the ID number "AUTO:". There are, for example, automatic failure types for

- Non-observance of the upper tolerance limit
- Non-observance of the lower tolerance limit
- Non-observance of action and warning limits

Failures the designation of which includes the number sign (#) automatically trigger the generation of a complaint if they are assigned to a measured value of one of the areas in-production inspection, goods receipt inspection or goods issue inspection.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure type, the system checks whether there is already a failure type with this key information.

The input of failure types is easy to handle. Only a failure analysis number and a corresponding designation have to be assigned.

Failure type groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure type. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The hierarchical tree entries of the group tree allow for the requested group to be selected and taken over. Then the "groups" field of the editing dialog for failure types shows the assigned group including the hierarchical group structure.

If failure types are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure type groups" application, which is described in the manual entitled "[MOC_Groups.pdf](#)".

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the failure type number as failure key.

By differentiating between active and inactive failure types, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure types at any time. Moreover, inactive failure types can be reactivated at any time.

Integration

The failure types catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected in measurement recording the failure catalogs help describe the deviations in more detail and represent it in a way that allows for analyses/reports to be performed. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition, this catalog is the basis for failure mode analyses relating to the failure types.

This catalog is also required for the creation of analysis selection catalogs as well as for using inspection chart characteristics within inspection planning and measurement recording. An analysis selection catalog includes a subset of all failures and restricts the list of failure (types) that can be selected during the collection process to those failures of the analysis selection catalog assigned to the characteristic. Consequently, the analysis selection catalog also determines the list of failure types for inspection chart characteristics.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure types of a group can be filtered in the "groups" tab using the icon  and selecting a failure type group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure types.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure types that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

12 Failure Locations

Summary

Menu	Master data → Quality management → Failure locations
Transaction code	floc
Function authorization	floc

The catalog of failure locations has been designed to describe the deviations occurred in more detail, e.g. deviations from specified limit values.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure location, the system checks whether there is already a failure location with this key information.

The input of failure locations is easy to handle. A failure analysis number and a corresponding designation only have to be assigned.

Failure location groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure location. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure then appears in the "groups" field of the editing dialog.

If failure locations are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure location groups" application, which is described in the manual entitled "[MOC_Groups.pdf](#)".

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the failure location number as failure key.

By differentiating between active and inactive failure locations, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure locations at any time. Moreover, inactive failure locations can be reactivated at any time.

Integration

The failure location catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected in measurement recording the failure catalogs help describe the deviation in more detail and represent it in a way that allows for analyses/reports to be performed. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition this catalog is the basis for failure mode analyses relating to the failure locations.

The failure locations of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failure (locations) that can be selected during the collection process to those failures of the analysis selection catalog that is assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure locations of a group can be filtered in the "groups" tab clicking the icon  and selecting a failure location group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure locations.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure locations that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of a tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

13 Failure Causes

Summary

Menu	Master data → Quality management → Failure causes
Transaction code	fcau
Function authorization	fcau

The catalog of failure causes has been designed to describe the occurred defects/failures in more detail, e.g. if limit values have been infringed.

Utilization

The "failure analysis number" field is the key field, i.e. when saving a new failure cause, the system checks whether there is already a failure cause with this key information.

The input of failure causes is easy to handle. A failure analysis number and a corresponding designation only have to be assigned.

Failure cause groups may optionally be defined beforehand. Consequently, the corresponding group can be assigned to the respective failure cause. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure then appears in the "groups" field of the editing dialog.

If failure causes are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "failure causes groups" application, which is described in the manual entitled "[MOC Groups.pdf](#)".

Under certain circumstances it might be reasonable and recommendable to use a self-explanatory structure for the failure cause number as failure key.

By differentiating between active and inactive failure causes, it can be defined whether or not they are still to be available in failure selection lists in the later data acquisition process. However, it is also possible to evaluate inactive failure causes at any time. Moreover, inactive failure causes can be reactivated at any time.

Integration

The failure cause catalog is used, among other things, within measurement recording and in complaint management. If deviations with respect to tolerance limits are detected in measurement recording the failure cause catalog helps describe the actual failure cause in more detail and represents it in a way that allows for analyses/reports to be made. Only in this way is it possible to determine failure mode analyses, take appropriate action (measures) and prevent the deviation from reoccurring.

In addition this catalog is the basis for failure mode analyses relating to the failure cause.

The failure causes of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failure (causes) that can be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Failure causes of a group can be filtered in the "groups" tab using the  icon and selecting a failure cause group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive failure causes.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies failure causes that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of a tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

14 Causers

Summary

Menu	Master data → Quality management → Originator
Transaction code	ori
Function authorization	ori

The catalog of causers/originators has been designed to describe the cause for an initial situation. In this context, the causer or originator does not imperatively have to be a person or a group of people. A machine can also be the causer/originator.

Utilization

The "causer number" field is the key field, i.e. while saving a new causer/originator, the system checks whether there is already a causer with this key information.

The input of causers is easy to handle. Only an ID number and a corresponding designation have to be assigned.

Causer groups may optionally be defined in advance and the corresponding group can be assigned to the originator. This option should not be missed out as it provides improved reports/evaluations. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for causers.

If causers are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "causer groups" application, which is described in the manual entitled [MOC Groups.pdf](#).

Under certain circumstances, it might be reasonable and recommendable to use a self-explanatory structure for the originator number as analysis key.

By distinguishing between active and inactive causers, it can be defined whether or not they are still to be available in selection lists for the causers in the later data acquisition process. However, it is possible to evaluate inactive originators at any time. Moreover, inactive causers can be reactivated at any time.

Integration

The catalog of originators is used, among other things, within measurement recording and in complaint management. A causer/originator may be added if defects are detected and documented within measurement recording. Only in this way is it possible to determine the basic causers, take appropriate action (measures) and prevent the defects from recurring during the failure mode analysis that follows.

The causers/originators of this catalog can also be integrated in analysis selection catalogs. An analysis selection catalog includes a subset of all failures and causers/originators and restricts the list of failures and causers that can be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Causers/originators of a group can be filtered in the "groups" tab using the icon  and selecting a causer group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive causers.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies causers that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

15 Measures

Summary

Menu	Master data → Quality management → Measures
Transaction code	meas
Function authorization	meas

The catalog of measures has been designed to define the action to be taken, for example, to prevent the detected failures from recurring or to allow for them to be identified in time.

Utilization

The "measure number" field is the key field, i.e. while saving a new measure, the system checks whether there is already a measure with this key information.

The input of measures is easy to handle. Only a measure number and a corresponding designation have to be assigned.

Measure groups may optionally be defined beforehand and the corresponding group can be assigned to the measure. This option should not be missed out as it provides improved reports/evaluations or structures. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for the measures.

If measures are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "measure groups" application, which is described in the manual entitled [MOC_Groups.pdf](#).

By distinguishing between active and inactive measures, it can be defined whether or not they are still to be available in selection lists for the measures in the later data acquisition process. Inactive measures can be reactivated at any time.

Integration

The measures catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected within measurement recording actions can be triggered immediately on the basis of the catalog of measures. At first the assigned measures have the status "open". In complaint management a party responsible (e.g. a person) as well as a deadline may be defined. In addition to this, an actual date, the completion in % and effectiveness in % may be specified as well. Normally, these fields are only filled out, once a measure has been completed (measure status switches to "done").

Subject to whether the corresponding license for the escalation management module has been purchased, the measures can also be sent to the defined person by e-mail.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Measures of a group can be filtered in the "groups" tab using the icon  and selecting a measure group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive measures.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies measures that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

16 Department

Overview

Menu	Master data → Quality management → Department
Transaction code	departm
Function authorization	departm

The catalog of departments has been designed to edit departments. Provided that there is an interface to a higher-level system (e.g. ERP system), departments can be created or changed automatically by interface.

Purpose

The “department” field uniquely identifies departments in all QM applications referring to the departments catalog. The departments catalog is used, in particular, by complaint management and when measures are created.

The “department” field is the key field, i.e. when a new department is saved it is checked whether or not a department already exists with this key information.

The differentiation between active and inactive departments determines whether or not these departments are still available in specific selection lists. For example, an inactive department cannot be assigned as being responsible for a measure. But reports on inactive departments are possible at any time and inactive departments can always be reactivated.

In addition to the department (number) and designation, a company (customer, supplier, manufacturer) may also be assigned. Furthermore, phone numbers, e-mail addresses etc. can be entered as well.

If a department is designated as being “responsible” this department is also included in the selection list for the “parties responsible”. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used in complaint management and when measures are created.

Integration

Department data is a global catalog used in many QM applications. The list below shows some of the applications referring to this catalog.

- External persons
- Customers
- Suppliers

- Manufacturers
- Complaint management
- Measures

Prerequisite

There are no special requirements.

Selection criteria

The company type and company number may be used as filter criteria in addition to the “department” and “designation” fields.

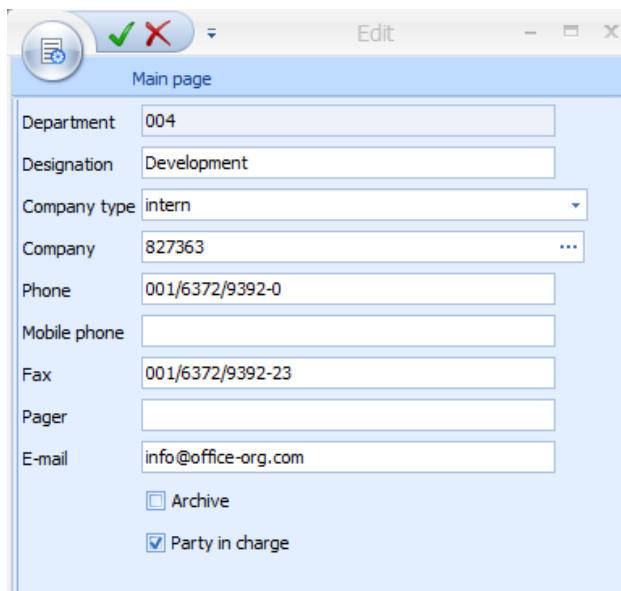
Using the filter field “inactive”, active or inactive departments can be selected.

Field descriptions

The fields are self-explanatory and not described in more detail for this reason.

Editing functions

This dialog opens for editing of a data record.



The screenshot shows a software window titled "Edit" with a standard Windows-style title bar. Inside the window, there's a section labeled "Main page". Below this, there are several input fields arranged vertically:

- Department: 004
- Designation: Development
- Company type: intern (dropdown menu)
- Company: 827363 (with a three-dot menu icon)
- Phone: 001/6372/9392-0
- Mobile phone: (empty)
- Fax: 001/6372/9392-23
- Pager: (empty)
- E-mail: info@office-org.com

At the bottom of the form, there are two checkboxes:

- ☐ Archive
- ☒ Party in charge

Toolbar

There are no other special function buttons besides the standard functions.

17 External Persons

Overview

Menu	Master data → Quality management → External persons
Transaction code	extper
Function authorization	extper

The catalog of “external persons” has been designed to define persons that are not edited in the HR master. Provided that there is an interface to a higher-level system (e.g. ERP system), external persons can be created or changed automatically by interface.

Purpose

The personal number uniquely identifies external persons in all QM applications referring to this catalog. The complaint management, in particular, uses the catalog of external persons.

The contact partners of customers, suppliers and manufacturers are defined in this catalog, for example. As the integrated HR master is not always available, it is also possible to include “internal” staff in this catalog. The same also applies for a licensed HR master. However, it is important that this catalog is not connected with the HR master and does not replace it. The catalog of external persons is currently only used in QM applications.

The “personal no.” field is the key field, i.e. when a new person is saved it is checked whether or not a person already exists with this key information.

The differentiation between active and inactive persons determines whether or not these persons are still available in specific selection lists. For example, an inactive person cannot be assigned as being responsible for a measure. But inactive persons can be reactivated at any time.

In addition to the personal number and designation, substantial address and contact data as well as the assignment to a company and department may be defined.

If a person is designated as being “responsible” this person is also included in the selection list for the “persons in charge”. Such selection lists are integrated in different detail applications. The list of parties responsible is mainly used in complaint management and when measures are created.

Integration

“External persons” is a global catalog used in many QM applications. The list below shows some of the applications referring to this catalog.

- Customers

- Manufacturers
- Suppliers
- Departments
- Measures
- Complaint management

Prerequisite

There are no special requirements.

Selection criteria

The last name, first name as well as the initial may be used as filter criteria in addition to the personal number.

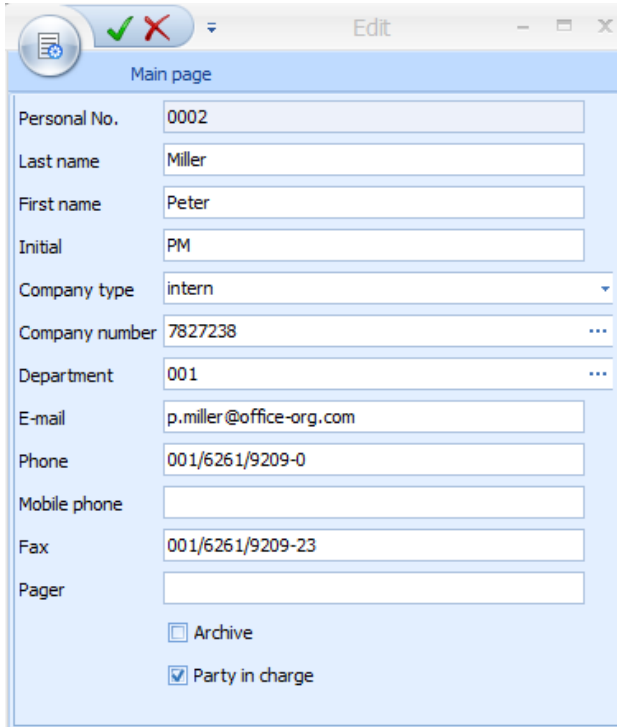
Using the filter field “inactive”, active or inactive persons can be selected.

Field descriptions

The fields are self-explanatory and not described in more detail for this reason.

Editing functions

This dialog opens for editing of a data record.



The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a toolbar with a document icon, a green checkmark, a red X, and a dropdown arrow. Below the toolbar is a tab labeled 'Main page'. The form contains the following fields and controls:

Personal No.	0002
Last name	Miller
First name	Peter
Initial	PM
Company type	intern
Company number	7827238 ...
Department	001 ...
E-mail	p.miller@office-org.com
Phone	001/6261/9209-0
Mobile phone	
Fax	001/6261/9209-23
Pager	

At the bottom of the form, there are two checkboxes:

- ☐ Archive
- ☒ Party in charge

Toolbar

There are no other special function buttons besides the standard functions.

18 Complaint Management

Overview

Menu	Quality management → Complaint management → Complaint management
Transaction code	cm
Function authorization	cm

The screenshot displays the 'Complaints management' application window. It features a top menu bar with 'Main page' and 'Profile'. A toolbar below the menu includes icons for data request, cancellation, printing, assignment, insertion, copying, editing, deletion, workflow, saving, and help. The main workspace is divided into three primary panes. The left pane, 'Selection', offers filters for complaint type, area, status, and date. The middle pane, 'Complaint', lists existing complaints with columns for type, area, ID, external number, receipt date, and recipient. The right pane, 'Complaint detail', provides a form to view or edit a specific complaint, including fields for area, type, ID, external number, receipt date, time, and recipient. Below the detail form is a 'Complaint detail' table and a 'Failure type' table for further analysis.

Utilization

This function allows for different types of complaints to be created. The types “customer complaint”, “supplier complaint” and “internal complaint” are provided by default. The complaint header should only include data that are not directly related to the item complained about. Details relating to the article/item are to be defined within the subordinate complaint details.

Integration

This application provides data for the following reports/evaluations:

- Failure analysis of complaints,
- Complaint analysis,

- Analysis of complaint costs and
- Measure tracking.

Different master data are used if a complaint is created, e.g.

- customers for a customer complaint,
- failures for the failure analysis
- measures for defining measures,
- costs for costs recording and
- inspection requirements for the assignment of a referenced inspection step.

Prerequisite

Relevant master data need to be edited/maintained to be able to create complaints. Which master data have to be maintained depends on the respective field of application. Normally, the master data

- Articles/items,
- Defects,
- Measures,
- Costs,
- Companies
- Departments and
- Staff

have to be maintained first.

Selection criteria

The sections that follow describe the selection criteria that are not self-explanatory.

Complaint tab

Complaint

The complaint number assigned manually or automatically may be filtered here.

Ext. complaint number

If the customer/supplier uses another complaint number than the number that is created for this complaint, it will be defined in the "ext. complaint number" field. This field may be filtered.

Complaining party tab

Complaining party type

The types "supplier", "customer", "department" and "person" may be selected. Subject to the selected type, an entry may be selected in the "complaining party" field.

Complaining party

The contents of the party in charge list may be filtered. Which entry is transferred to the list of responsible parties is defined within the master data.

Designation

The content of the field "name 1" of the list of responsible parties is filtered. This is the name of the department for departments, the last name for external persons and the company name for companies.

Contact partners tab**Contact partner type**

Different types may be selected. An entry may be selected in the "contact" field, subject to the selected type.

Contact

The contents of the party in charge list may be filtered. Which entry is transferred to the list of responsible parties is defined within the master data.

Designation

The content of the field "name 1" of the list of responsible persons may be filtered. This is the department name for departments, the last name for external persons and the company name for companies.

Party in charge tab**Party in charge type**

Different types may be selected. An entry may be selected in the "party in charge type" field, subject to the selected type.

Party in charge type

The contents of the party in charge list may be filtered. Which entry is transferred to the list of responsible parties is defined within the master data.

Designation

The content of the field "name 1" of the list of responsible persons may be filtered. This is the department name for departments, the last name for external persons and the company name for companies.

Field descriptions

Complaint tab

Area

Selection list of the configured areas

Complaint

In case a complaint number has not been assigned manually, it will be assigned automatically, once HYDRA has been saved. In combination with the area HYDRA creates a unique complaint number.

Type of complaint

Customer complaint, supplier complaint or internal complaint.

Ext. complaint number

If the customer/supplier uses another complaint number than the number that is created for this complaint, it will be defined in the "ext. complaint number" field.

Received by

Once saved, the registered user is entered in this field.

Date of receipt/time

The current system date/time will be entered upon saving.

Status

List of configured complaint statuses

Result

List of configured complaint results

Target date, time

Information field if a corresponding specification is to be entered for dealing with the complaint.

Actual date/time

A date/time when the complaint is to be considered as "done" may be entered here. The actual status is not checked in this context.

Complaining party type

Different types may be selected. The complaint type is not checked. The type "customer" should be entered here if it is a customer complaint to be able to enter a customer as the complaining party.

Complaining party

The contents of the party in charge list may be filtered. Which entry is transferred to the list of responsible parties is defined within the master data. The selected entry is used as the complaining party.

Complaining party name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the complaining party are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Contact partner type

Different types may be selected.

Contact

The list of responsible parties is displayed. Which entry is transferred to the list of responsible parties is defined within master data. The selected entry is used as the contact.

Contact name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the contact are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Party in charge type

Different types may be selected. This field does not have a special function, i.e. it depends on the corresponding application specifying who is to be entered as the responsible party. Normally, the person selected here is generally responsible for the entire complaint. However, this is not monitored by special functions.

Party in charge

The list of responsible parties is displayed. Which entry is transferred to the list of responsible parties is defined within master data. The selected entry is used as the responsible party.

Party in charge name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the responsible party are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Additional data tab**Cost center**

Information field to specify a cost center

Delivery note

Information field to state a delivery note number. This field is not checked against the previously selected complaint type (e.g. supplier complaint).

Delivery date

Information field to state a delivery date. This field is not checked against the previously selected complaint type (e.g. supplier complaint).

Storage location

Information field to specify a storage location.

Toolbar**Copy a complaint header**

Function authorization: cm.insert

The copy function opens the selected complaint and also allows for the key fields to be changed. But only the complaint header is copied upon saving. The complaint details, i.e. documents, measures, costs and failure analyses that might pertain to the "copy template" are not copied.

**Referencing of complaints**

Function authorization: none

A list/application is opened that references complaints (e.g. did a supplier complaint result from a customer complaint) or shows complaints that have already been referenced.

**Calling the workflow history**

Function authorization: cmwf.edit

Opens the graphic view of the referenced workflow for the complaint header. The graphic states the current processing status. In addition to the graphic, the list also shows every action pertaining to the workflow.

Detail application "complaint detail"

Function authorization	cmd
------------------------	-----

Details		Additional data		User fields		Administration	
Details							
Complaint	002			Detail no.	1		
Id	23 9383 712			Article designation	Headlight XA 77		
Drawing issue number	B						
Supplier							
Supplier No.	928762			Supplier designation	SANYO Co. Ltd.		
Purchase order number							
Serial number							
Batch							
Status							
Status				Result			
Party in charge							
Party in charge type	ext. Pers.						
Party in charge	0219						
Party in charge name 1	Miles						
Party in charge name 2	Betty						
Party in charge name 3	OVERHEAD						
Team				Team name			

All pieces of information that are directly connected to the item/material complained about should be defined within the complaint details, as the item complained about is only assigned at this point.

This allows for "collective complaints" to be recorded. Consequently, a customer can complain about different items/articles e.g. using one complaint number. A complaint detail may be created and analyzed separately for the complaint header of each article/item complained about. It is also possible to create several complaint details for one article/item. This is required, for example, if different batches, etc. are complained about and need to be analyzed separately.

The field "complaint" is assigned to the complaint number of the currently selected complaint, when editing or creating new complaint details. The numeric value of the "detail" field is generated automatically, once this new data record has been saved and cannot be changed anymore.

Field descriptions

"Details" tab

Complaint

Shows the complaint number of the superordinate complaint.

Detail no.

The complaint detail number is shown here when editing. This field is empty when a new data record is created and is assigned automatically to a complaint number that is unique within this complaint upon saving.

Article number, article designation, drawing issue number

The selection list of article master data may be opened from which an article/item can be selected to specify the article/item complained about. The corresponding article designation is shown within the complaint details. Instead of selecting the article, the article number as well as drawing issue number may also directly be entered. In this case, the article designation is determined from the master data catalog and displayed upon saving.

Supplier no.

Direct input of the supplier number or selection by opening the supplier catalog. The supplier name is also displayed after saving.

Purchase order number

Information field to enter a purchase order number. This field input does not depend on the selected complaint type.

Serial number

Information field to enter a serial number.

Batch

Information field to enter a batch.

Status

Selects/shows the complaint detail status

Result

Selects/shows the result of the complaint detail

Party in charge type

Different types may be selected when inputting data. This field does not have a special function, i.e. it depends on the corresponding application specifying who is to be entered as the responsible party. Normally, the person selected here is generally responsible for dealing with the complaint in more detail, i.e. provision of information about the article/item complained about. However, this is not monitored by special functions.

Party in charge

The list of responsible parties or the assigned parties in charge are displayed. Which entry is transferred to the list of responsible parties is defined within master data. The selected entry is used as the responsible party.

Party in charge name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the responsible party are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Team

A team may be entered or selected. Teams are defined within master data. After saving, the designation is displayed in addition to the team number.

Additional data tab**Delivery value**

Information field to enter/display a delivery value. This field is available irrespective of the complaint type.

Complaint value

Information field to enter/show the complaint value.

Delivery quantity

Information field to enter/display a delivery quantity. This field is available irrespective of the complaint type.

Complaint quantity

Information field to enter/show the quantity specified in the complaint.

Share of the complaint

Information field to enter/show the proportion of the complaint. The content of this field is not calculated automatically.

Checked quantity

Information field to enter/show the checked quantity. The customer has to decide whether this refers to the quantity checked by the customer or by the complaining party.

Defective quantity

Information field to enter/show the faulty quantity. The customer has to decide whether this refers to the defective quantity identified by the customer or by the complaining party.

Share of defects

Information field to enter/show the share of defects. The content of this field is not calculated automatically.

Inspection requirement 1

Shows the assigned inspection requirement or provides the option to choose an inspection requirement from the list of inspection requirements. The selection list provides all inspection requirements from all sectors. The inspection requirement number is displayed in this field, once an inspection requirement has been taken over. In addition to this, the corresponding sector is shown, e.g. "E" for goods receipt or "F" for production.

Inspection requirement 2

Shows the second inspection requirement assigned or provides the option to choose an inspection requirement from the list of inspection requirements. The selection list provides all inspection requirements from all sectors. The inspection requirement number is displayed in this field, once an inspection requirement has been taken over. In addition to this, the corresponding sector is shown, e.g. "E" for goods receipt or "F" for production.

Toolbar



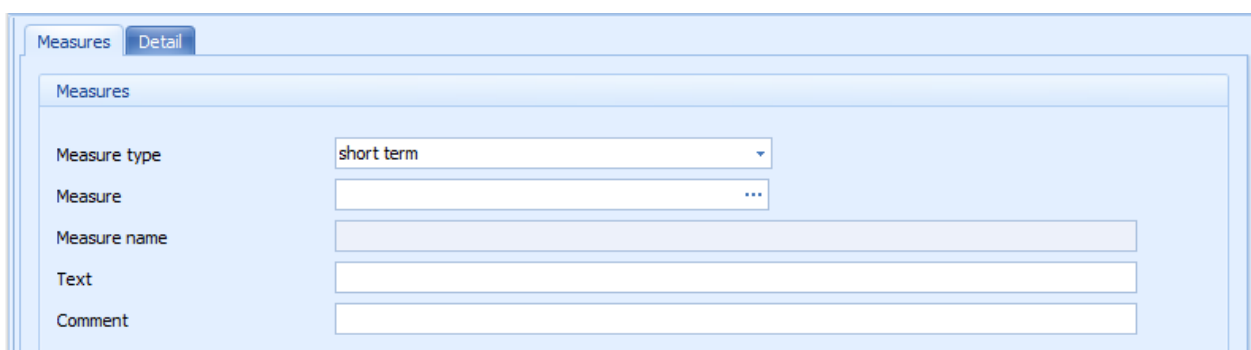
Calling the workflow history

Function authorization: cmdwf.edit

Opens the graphic view of the workflow referenced for the complaint detail. The graphic states the current processing status. In addition to the graphic, a list also shows every action pertaining to the workflow.

Detail application "measures in the complaint header and complaint details as well as in the failure analysis"

Function authorization	cmme for measures in the complaint header cmdme for measures in the complaint detail cmdfa for measures in the failure analysis
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Measures			
Status			
Status			
Fulfillment [%]		Effectiveness [%]	
☐ External			
Party in charge			
Party in charge type			
Party in charge	...		
Name 1			
Name 2			
Name 3			
Date			
Target date		Target time	
Actual date		Actual time	

A list of assigned measures is available for each selected complaint or complaint detail. Further measures may be added or existing measures may be changed, complemented or deleted at any time. The measures created here are also included in the "measures tracking" application where they may also be edited.

When a new data record is created, a "measure" needs to be indicated to be able to save the data record. In addition to the field "measure", it is also possible to open the "measures" master data catalog from which a measure may be chosen and assigned. As an alternative to selecting measures, measures may also be entered directly.

The statuses "open", "read", "in process" and "completed" are provided by default. In addition to the measure type "no assignment", the types "short-term", "medium-term" and "long-term" may also be selected. These statuses can be enhanced according to the customer's requirements by customizing the system.

Subject to the selected "party in charge type", the pre-filtered list of responsible parties is opened by clicking the magnifying glasses button. Entries that have been assigned the flag "responsible" in the relevant master data are shown only. The list of the types that may be selected matches the types that have already been defined for the complaint header.

It is important to specify a target date. Based on this information, corresponding filters can be set in the measures tracking function.

The fields "fulfillment" and "effectiveness" have been designed to finally "assess" the defined measure.

Field descriptions

Measures tab

Measure type

The available measure types are displayed or can be chosen. The types "short-term", "medium-term", "long-term" and "no assignment" are provided by default.

Measure

The measure number is shown or can be selected or it may also be entered directly. The relevant master data catalog can be opened for selection purposes.

Measure designation

The designation of the assigned measure number is shown. If the measure number is input directly, the designation will only be shown upon saving.

Text

Free text field to enter a complementary measure text

Comment

Free text field to enter a complementary comment for the measure.

Detail tab

Status

Available measure types can be displayed or selected. The types "in process", "read", "done" and "open" are provided by default.

Fulfillment [%]

Fulfillment in % can be displayed or entered.

Effectiveness [%]

Effectiveness in % can be displayed or entered.

External

This field can be used to control the printout of forms in future. However, it does not have a special function.

Party in charge type

Different types may be selected when inputting data. The type "external person" is used, as normally people are responsible for dealing with measures.

Party in charge

The party in charge is shown or it may be chosen from the list of responsible parties. Which entry is transferred to the list of responsible parties is defined within master data. The selected entry is used as the responsible party.

Party in charge name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the responsible party are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Target date/time

A date and optionally a time by which the measure has to be finished may be displayed or entered. It will not be monitored automatically, whether or not this time limit is kept. The content of this field is fundamental to the "measures tracking" application, as it may be determined manually for all measures (global) which of them have exceeded the target date, for example.

Actual date/time

A date and optionally a time specifying when the measure was finished may be displayed or entered. However, this field does not have a special function. In the "measures tracking" application, this field can be used to determine manually which measures have been completed with a delay.

Detail application "costs in the complaint header and complaint detail"

Function authorization	cmco for the costs in the complaint header cmdco for the costs in the complaint detail
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The screenshot shows a software window titled 'Costs'. Inside, there is a sub-section also titled 'Costs'. Below this, there are five input fields with labels to their left: 'Cost no.' (with a dropdown arrow), 'Cost des.' (a long text box), 'Duration' (with a unit selector set to 'Hours'), 'Amount' (a text box), and 'Comment' (a long text box).

The list of assigned costs is available for each selected complaint or complaint detail. Further costs may be added or existing costs may be changed, complemented or deleted at any time. The costs defined here can be evaluated in the report "analysis of complaint costs".

When a new data record is created, a "cost type" needs to be indicated to be able to save the data record. In addition to the field "cost number", it is also possible to open the "cost types" master data catalog from which a cost type may be chosen and assigned. Costs may also be entered directly as an alternative to them being selected.

Provided that an initial duration ("init. duration" field) and an amount record ("cost rate amount" field) have been assigned in the cost types catalog of the cost type, the field "duration" is assigned to the value from "init. duration" and the field "amount" is assigned to the product from "init duration" and "cost rate amount" in the dialog for complaint costs. If the field "duration" is changed or taken over unchanged as a part of creating a new data record, the "amount" field will be recalculated automatically, once the "duration" field has been saved. The amount is only calculated once as a part of initial data creation. The fields "duration" and "amount" have to be changed manually if they need to be changed after the initial data creation. As they are not recalculated automatically.

Field descriptions

Costs tab

Cost no.

The cost number is shown or can be selected or it may also be entered directly. The relevant master data catalog can be opened for selection purposes.

Cost designation

The designation of the assigned cost number is shown. If the cost number is directly input the designation will only be shown upon saving.

Duration

The duration is entered or displayed in the format "hh:mm:ss". When transferring a cost type from master data, this field is initially assigned to the master data field "init. duration".

Amount

The (calculated) amount is entered or displayed. When taking over a cost type from master data, this field is initially assigned to the master data field "init. amount". As a part of initial data creation, the value of this field is calculated by multiplying the "duration" field with the original value entered in this field upon saving.

Detail application "documents in the complaint header and complaint details as well as in the failure analysis"

Function authorization	cm for documents in the complaint header cmd for documents in the complaint details cmdfa for documents in the failure analysis
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The screenshot shows a window titled 'Main page' with a toolbar at the top containing a green checkmark, a red X, and a dropdown arrow. The window is divided into two main sections. The left section contains labels for 'Designation', 'Type', and 'Text'. The right section contains the corresponding input fields: a text box with 'Failure description' for Designation, a checkbox labeled 'External' for Type, and a large text area with the placeholder text 'Text field for a comprehensive description' for Text.

The screenshot shows the same 'Main page' window, but with the 'Type' dropdown set to 'URL'. The 'Designation' text box now contains the word 'Picture'. The 'Text' label is no longer visible. A new 'File name/address' label and text box have appeared, containing the path '\\complaint\23\pictures\23-1.jpg'. A small ellipsis button is visible to the right of the text box.

Provided that the "documents" tab has been enabled, as many documents as required may be assigned to each complaint header, complaint detail and failure analysis. By enabling these tabs, the toolbar offers corresponding buttons to edit the documents. The documents that have already been assigned can be viewed in a list in the mentioned tab.

When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types "FILE", "URL" and "Text" are provided. The file name including path may be entered manually with the "file" type. The "URL" file type allows access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Once saved, a consecutive item number (numeric) is automatically assigned to each entered document. In addition to this, the "external" checkbox specifies for the failure analysis whether or not the document is to be part of an 8D report. Finally, the 8D report determines whether or not this field is to be filtered at all.

Toolbar

In addition to the standard functions, there is also a button to show the documents.



Show documents

If a document link is defined, this button opens and shows this document. However, a program, which can show the linked file type, has to be installed on the PC. HYDRA paths are to be configured accordingly to open the documents.

Detail application "failure type" (failure analysis)

Function authorization	cmdfa for failure types in the failure analysis
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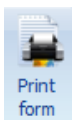
As many failure types as required may be assigned to each complaint detail. A failure type needs to be assigned to print out 8D reports. The list of failure types has to be opened to assign a failure type. The required failure type can directly be entered or chosen from a master data catalog. If the failure number is entered directly, the corresponding designation will only be shown after saving.

In addition to this, a comment may also be assigned.

The specified weighting affects failure analysis of complaint management. If the error occurs 10 times, i.e. 10 items are defective in this respect; the relevant value should be entered here.

Toolbar

In addition to the standard functions, the function for the printing of forms (8D report) is also available.



Show documents

Function authorization: cmdfa.print

This function opens the detail application "printing", which in this case enables printing of the 8D report. The 8D report only includes contents referring to the selected failure type and that are assigned to the "external" flag.

"Print" detail application

Function authorization	cmdfa.print for printing the failure analysis (8D report)
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The print dialog of the failure type opens a list of available reports. By default, printing is restricted to the 8D report. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the contents of the list of forms of the corresponding print dialog, are defined within the master data of quality management. The basis for new forms is established and the corresponding form properties are defined there. A corresponding license is required to be able to change the forms with respect to content and design.

Print - toolbar

There are no other special function buttons in addition to the standard functions/features.

Detail applications "failure location", "failure cause", "causer" of the failure type

Function authorization	cmdfa for failure locations, failure causes, causers in the failure analysis
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As many failure locations, failure causes and causers as required may be assigned to every failure type. Several tabs are displayed.

A failure type has to be selected beforehand to be able to perform the assignment. Subject to the selected tab, the relevant entry may be created using the toolbar.

The creation process corresponds to the functions described in the detail application "failure type" (failure analysis).

Only data assigned to the flag "external" are printed in the 8D report.

Detail applications "measure" for the failure type

Function authorization	cmdfa for measures in the failure analysis
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As many measures as required may be assigned to each failure type.

A failure type has to be selected beforehand to enable assignment.

The creation of a measure corresponds to the functions described in the detail application "measures in the complaint header and complaint detail as well as the failure analysis".

Only data assigned to the flag "external" are printed in the 8D report.

Detail application "documents of the failure type"

Function authorization	cmdfa for measures in the failure analysis
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As many documents as required may be assigned to each failure type.

A failure type has to be selected beforehand to enable assignment.

The creation of a document corresponds to the functions described in the detail application "documents in the complaint header and complaint detail as well as the failure analysis". In addition to this, an assignment category may also be indicated. This is important to the 8D report. By default, the categories "comment", "forecast", "control of success" and "no assignment" are available.

Only data assigned to the flag "external" are printed in the 8D report.